

Homework Assignment 1

1. **(14 points)** Solve the inequality $\frac{1}{4} < \left| \frac{2x-1}{x} \right| \leq \frac{1}{2}$ in terms of intervals and illustrate the solution set on the real number line.
2. **(14 points)** Sketch the set $\left\{ (x, y) \in \mathbb{R}^2 : \sqrt{2x^2 + 16x + 34} + 2 \leq y \leq -|x + 4| + 5 \right\}$ in the xy -plane.
3. **(14 points)** Recall that we use the symbol $\parallel$ when two lines are parallel or the same, and the symbol $\perp$ when two lines are perpendicular. Let l, r, s be non-vertical lines. Show that:
 - (a) If $l \parallel r$ and $r \parallel s$, then $l \parallel s$.
 - (b) If $l \parallel r$ and $r \perp s$, then $l \perp s$.
 - (c) If $l \perp r$ and $r \perp s$, then $l \parallel s$.
4. **(14 points)** Find the line perpendicular to the line $2x - \sqrt{5}y = 6 + 4\sqrt{5}$ that passes through the intersection of the conics $4x^2 + 5y^2 - 24x + 40y + 96 = 0$ and $\frac{(y+4)^2}{4} - \frac{(x-3)^2}{5} = 1$, which is the closest to the origin.
5. **(14 points)** Find all real numbers t such that $\cos t = \sin^2 t - \frac{1}{4}$.
6. **(16 points)** Let c be a fixed real number. Find the domain, the range, and sketch the following function.
$$f(x) = \begin{cases} c \left(\frac{x^3 + 3x^2 + 3x + 1}{x + 1} \right) & \text{if } x < -\frac{\pi}{6}, \\ 2 \cos x - \sqrt{3} & \text{if } -\frac{\pi}{6} \leq x \leq 3\pi, \\ \frac{x^2 - 6\pi x + 9\pi^2}{x - 3\pi} - 2 - \sqrt{3} & \text{if } 3\pi < x < 5\pi, \\ e^{-\frac{x}{5\pi}} & \text{if } x \geq 5\pi. \end{cases}$$
7. **(14 points)**
 - (a) Show that, if $0 < a < b$ and $0 < c < d$ then $ac < bd$.
 - (b) Using part a), show that if $0 < x < y$ then $x^3 < y^3$.